



CS464 - Data and Network Security

Week 02

Hafiz Syed Ahmed Qasim

ahmed.qasim@giki.edu.pk



Introduction to Symmetric Encryption

What is Symmetric Encryption (1/2)

- 1 Symmetric encryption was the only type of encryption in use before the 1970s when public-key encryption was developed.
- 2 It remains by far the most widely used type of encryption today.
- 3 This lecture examines symmetric ciphers and provides a general model for the symmetric encryption process.
- 4 The model helps understand the context within which encryption algorithms are used.

What is Symmetric Encryption (2/2)

- 1 Later lectures will introduce the two most widely used symmetric ciphers: DES and AES.
- 2 The symmetric encryption process involves converting plaintext to ciphertext using a secret key.
- 3 Both sender and receiver must share the same secret key for the system to work.
- 4 The lecture also examines algorithms used before the computer era and briefly looks at steganography.

Key Terms and Definitions (1/2)

1

An original message is known as the plaintext, while the coded message is called the ciphertext.

2

The process of converting from plaintext to ciphertext is known as enciphering or encryption.

3

Restoring the plaintext from the ciphertext is called deciphering or decryption.

4

The many schemes used for encryption constitute the area of study known as cryptography.

Key Terms and Definitions (2/2)

- 1 Such a scheme is known as a cryptographic system or a cipher.
- 2 Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis.
- 3 Cryptanalysis is what the layperson calls "breaking the code."
- 4 The areas of cryptography and cryptanalysis together are called cryptology.



Symmetric Cipher Model

Section 3.1

Five Ingredients of Symmetric Encryption (1/2)

- 1** The first ingredient is plaintext, which is the original intelligible message or data fed into the algorithm as input.
- 2** The second ingredient is the encryption algorithm, which performs various substitutions and transformations on the plaintext.
- 3** The third ingredient is the secret key, which is also input to the encryption algorithm.
The key is a value independent of the plaintext and of the algorithm.

Five Ingredients of Symmetric Encryption (2/2)



The algorithm will produce a different output depending on the specific key being used at the time.

4

The fourth ingredient is ciphertext, which is the scrambled message produced as output.

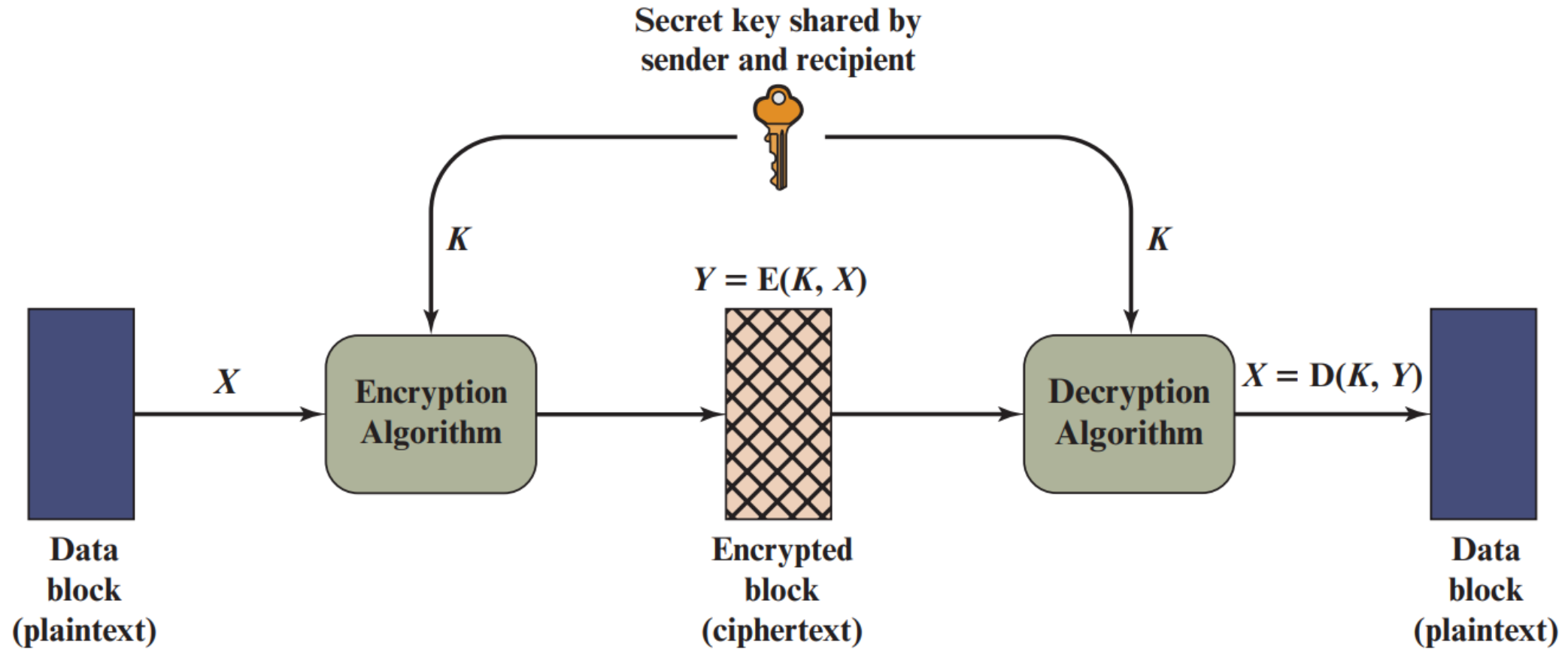
The ciphertext depends on the plaintext and the secret key.

5

The fifth ingredient is the decryption algorithm, which is essentially the encryption algorithm run in reverse.

It takes the ciphertext and the secret key and produces the original plaintext.

Simplified Model of Symmetric Encryption



Two Requirements for Secure Use (1/2)

1

The first requirement is that we need a strong encryption algorithm.

At a minimum, we would like the algorithm to be such that an opponent who knows the algorithm and has access to one or more ciphertexts would be unable to decipher the ciphertext or figure out the key.



This requirement is usually stated in a stronger form.

The opponent should be unable to decrypt ciphertext or discover the key even if he or she is in possession of a number of ciphertexts together with the plaintext that produced each ciphertext.

Two Requirements for Secure Use (2/2)

2

The sender and receiver must have obtained copies of the secret key in a secure fashion.

They must also keep the key secure.



If someone can discover the key and knows the algorithm, all communication using this key is readable.



We assume that it is impractical to decrypt a message based on ciphertext plus knowledge of the encryption and decryption algorithm.

We do not need to keep the algorithm secret, only the key must remain secret.

The Encryption Process (1/2)



A source produces a message in plaintext, $X = [X_1, X_2, \dots, X_M]$.

The M elements of X are letters in some finite alphabet.



Traditionally, the alphabet usually consisted of the 26 capital letters.

Nowadays, the binary alphabet $\{0, 1\}$ is typically used.



For encryption, a key of the form $K = [K_1, K_2, \dots, K_J]$ is generated.

If the key is generated at the message source, it must be provided to the destination via a secure channel.

The Encryption Process (2/2)



With the message X and the encryption key K as input, the encryption algorithm forms the ciphertext $Y = [Y_1, Y_2, \dots, Y_N]$.

We can write this as $Y = E(K, X)$.



This notation indicates that Y is produced by using encryption algorithm E as a function of the plaintext X .

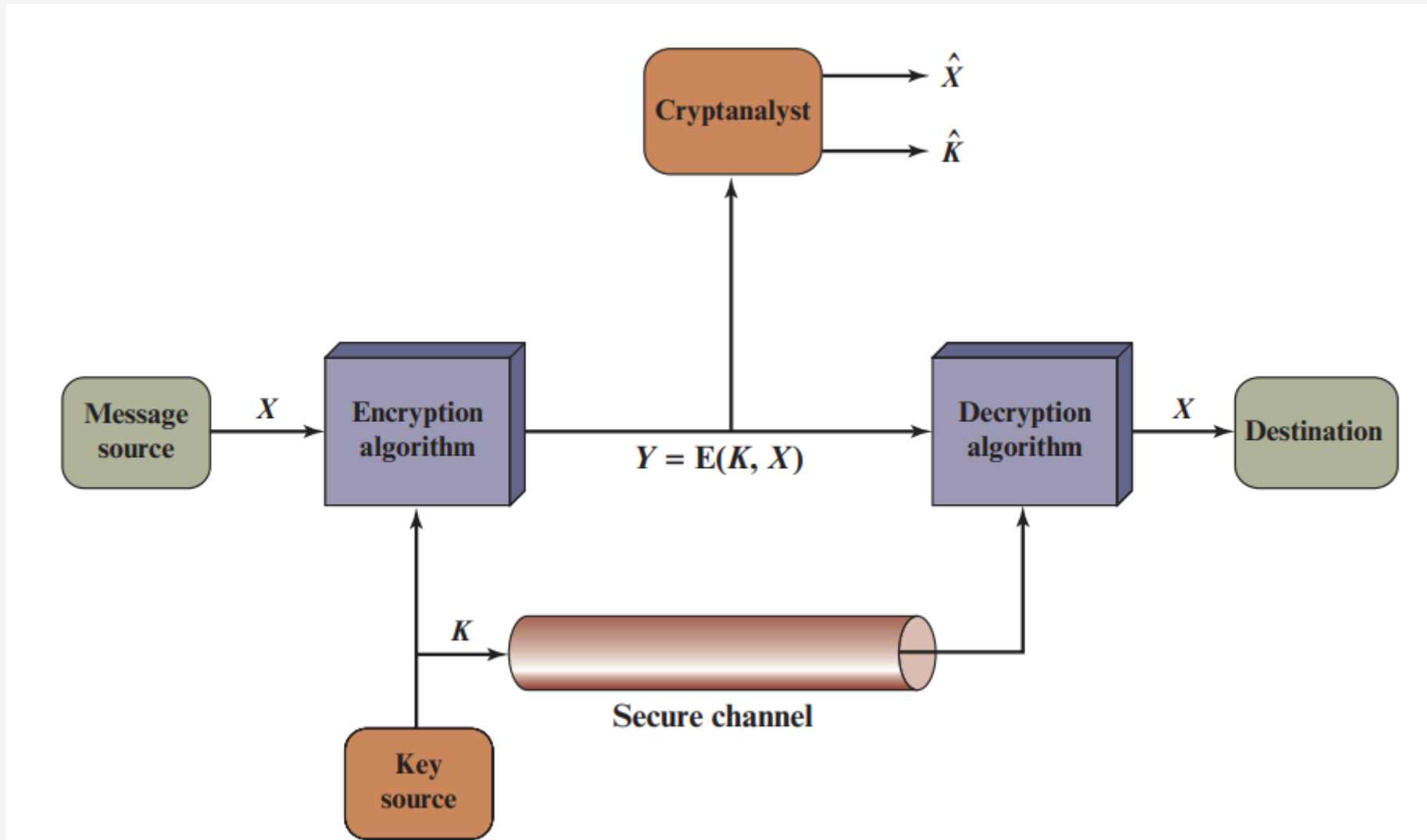
The specific function is determined by the value of the key K .

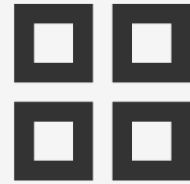


The intended receiver, in possession of the key, is able to invert the transformation: $X = D(K, Y)$.

An opponent, observing Y but not having access to K or X , may attempt to recover X or K or both.

Model of Symmetric Cryptosystem





Cryptography - Three Dimensions

Understanding the Classification of Cryptographic Systems

First Dimension: Type of Operations (1/2)



Cryptographic systems are characterized along three independent dimensions.

The first dimension is the type of operations used for transforming plaintext to ciphertext.



All encryption algorithms are based on two general principles: substitution and transposition.

Substitution maps each element in the plaintext into another element.

Transposition rearranges elements in the plaintext.



The fundamental requirement is that no information be lost during these operations.

All operations must be reversible.

First Dimension: Type of Operations (2/2)



Substitution involves mapping each element in the plaintext into another element.

The elements can be bits, letters, or groups of bits or letters.



Transposition involves rearranging elements in the plaintext.

Most systems referred to as product systems involve multiple stages of substitutions and transpositions.



The fundamental requirement is that no information be lost during these operations.

This ensures that all operations are reversible.

These two principles form the foundation of all encryption algorithms.

Second Dimension: Number of Keys (1/2)



The second dimension is the number of keys used in the cryptographic system.

If both sender and receiver use the same key, the system is referred to as symmetric encryption.



Symmetric encryption is also called single-key encryption or secret-key encryption.

It is also known as conventional encryption.



In symmetric encryption, both parties share the same secret key.

This key is used for both encryption and decryption processes.

Second Dimension: Number of Keys (2/2)



If the sender and receiver use different keys, the system is referred to as asymmetric encryption.

Asymmetric encryption is also called two-key encryption or public-key encryption.



In asymmetric encryption, each participant has a pair of keys.

One key is public and can be freely distributed.

The other key is private and must be kept secret by the owner.



This distinction between symmetric and asymmetric encryption is fundamental to cryptography.

The choice between them depends on the specific application requirements.

Third Dimension: Processing Method (1/2)



The third dimension is the way in which the plaintext is processed.

A block cipher processes the input one block of elements at a time.



It produces an output block for each input block.

The block cipher works on fixed-size blocks of plaintext data.



Each block is encrypted independently of the others.

This makes block ciphers suitable for data at rest and file encryption.

Block ciphers are the most common type of symmetric cipher in use today.

Third Dimension: Processing Method (2/2)



A stream cipher processes the input elements continuously.

It produces output one element at a time as it goes along.



Stream ciphers work on individual bits or bytes of plaintext data.

Each element is encrypted as it arrives, making stream ciphers ideal for streaming applications.



Stream ciphers are typically used for real-time communications.

They are often implemented as XOR operations with a keystream.

Both block ciphers and stream ciphers have their advantages and use cases.



Cryptanalysis and Brute-Force Attack

Understanding Encryption Security and Vulnerabilities

Two Approaches to Attack (1/2)



Typically, the objective of attacking an encryption system is to recover the key in use.

The goal is not simply to recover the plaintext of a single ciphertext.



There are two general approaches to attacking a conventional encryption scheme.



The first approach is cryptanalysis.

Cryptanalytic attacks rely on the nature of the algorithm.

They also rely on perhaps some knowledge of the general characteristics of the plaintext.

Sometimes cryptanalysis uses knowledge of sample plaintext-ciphertext pairs.

Two Approaches to Attack (2/2)



The second approach is a brute-force attack.

The attacker tries every possible key on a piece of ciphertext.

The goal is to obtain an intelligible translation into plaintext.



On average, half of all possible keys must be tried to achieve success.



If either type of attack succeeds in deducing the key, the effect is catastrophic.

All future and past messages encrypted with that key are compromised.

This makes key security essential for maintaining the confidentiality of communications over time.



Types of Cryptanalytic Attacks

Classification of Attack Methods Based on Available Information

Ciphertext-Only Attack (1/2)



The most difficult problem is presented when all that is available is the ciphertext only.
In some cases, not even the encryption algorithm is known.



In general, we can assume that the opponent does know the algorithm used for encryption.



One possible attack is the brute-force approach of trying all possible keys.
If the key space is very large, this becomes impractical.

Ciphertext-Only Attack (2/2)



The opponent must have some general idea of the type of plaintext that is concealed.

For example, the plaintext might be English or French text.

It could be an EXE file or a Java source listing.



The ciphertext-only attack is the easiest to defend against.

This is because the opponent has the least amount of information to work with.



The analyst must make educated guesses about the plaintext content based on statistical analysis of the ciphertext patterns and frequencies.

Known-Plaintext Attack (1/2)



In many cases, the analyst has more information than just the ciphertext.

The analyst may be able to capture one or more plaintext messages and their encryptions.



The analyst may know that certain plaintext patterns will appear in a message.

For example, a file encoded in Postscript format always begins with the same pattern.

There may be a standardized header or banner to an electronic funds transfer message.



All these are examples of known plaintext.

With this knowledge, the analyst may be able to deduce the key based on the way the known plaintext is transformed.

Known-Plaintext Attack (2/2)



Closely related to the known-plaintext attack is what might be referred to as a probable-word attack.

If the opponent is working with the encryption of some general prose message, he or she may have little knowledge of what is in the message.



However, if the opponent is after some very specific information, then parts of the message may be known.



For example, if an entire accounting file is being transmitted, the opponent may know the placement of certain key words in the header of the file.

As another example, the source code for a program developed by Corporation X might include a copyright statement in some standardized position.

Chosen-Plaintext Attack (1/2)



If the analyst is able somehow to get the source system to insert into the system a message chosen by the analyst, then a chosen-plaintext attack is possible.



In general, if the analyst is able to choose the messages to encrypt, the analyst may deliberately pick patterns.

These patterns can be expected to reveal the structure of the key.



The analyst designs messages that will expose weaknesses in the encryption algorithm.

The analyst then analyzes the resulting ciphertext to determine information about the key or the algorithm itself.

Chosen-Plaintext Attack (2/2)



The chosen-plaintext attack gives the analyst significant power.

The analyst can systematically probe the encryption system by choosing specific inputs and observing how the encryption algorithm transforms the plaintext.



This allows the analyst to discover relationships between the plaintext and ciphertext.

Such attacks are more powerful than ciphertext-only attacks.



They require the analyst to have some level of access to the encryption system.

However, the rewards can be substantial if the access can be obtained.

The analyst can potentially recover the key through carefully designed chosen-plaintext attacks.

Chosen-Ciphertext and Chosen-Text Attacks (1/2)



Table 3.1 lists two other types of attack: chosen ciphertext and chosen text.

These are less commonly employed as cryptanalytic techniques.

However, they are nevertheless possible avenues of attack.



In a chosen-ciphertext attack, the analyst can choose ciphertext to be decrypted.

The analyst then analyzes the resulting plaintext.

This can reveal information about the decryption process.



A chosen-text attack combines both chosen-plaintext and chosen-ciphertext capabilities.

The analyst can influence both the encryption and decryption processes.

Chosen-Ciphertext and Chosen-Text Attacks (2/2)



These attacks require the analyst to have access to both encryption and decryption capabilities.

The analyst can control both the input plaintext and the ciphertext to be decrypted.

This gives the analyst maximum information about the system.



Such attacks are rare in practice because they require significant access to the encryption system.

However, they are theoretically powerful.



Only relatively weak algorithms fail to withstand a ciphertext-only attack.

Generally, an encryption algorithm is designed to withstand a known-plaintext attack.

More robust algorithms aim to resist chosen-plaintext and chosen-ciphertext attacks as well.

Types of Attacks on Encrypted Messages

Type of Attack	Known to Cryptanalyst
Ciphertext Only	■ Encryption algorithm ■ Ciphertext
Known Plaintext	■ Encryption algorithm ■ Ciphertext ■ One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	■ Encryption algorithm ■ Ciphertext ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	■ Encryption algorithm ■ Ciphertext ■ Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key ■ Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Security Concepts: Unconditional vs Computational (1/2)



Two more definitions are worthy of note.

An encryption scheme is unconditionally secure if the ciphertext generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext.

This is true no matter how much ciphertext is available.



That is, no matter how much time an opponent has, it is impossible for him or her to decrypt the ciphertext.

The required information is simply not there.

With the exception of a scheme known as the one-time pad, there is no encryption algorithm that is unconditionally secure.

Security Concepts: Unconditional vs Computational (2/2)



Therefore, all that the users of an encryption algorithm can strive for is an algorithm that meets one or both of the following criteria.



The first criterion is that the cost of breaking the cipher exceeds the value of the encrypted information.



The second criterion is that the time required to break the cipher exceeds the useful lifetime of the information.



An encryption scheme is said to be computationally secure if either of the foregoing two criteria are met.

Unfortunately, it is very difficult to estimate the amount of effort required to cryptanalyze ciphertext successfully.

This makes it challenging to determine computational security with certainty.

Strong Encryption Properties (1/2)



For users, security managers, and organization executives, there is a requirement for strong encryption to protect data.



The term strong encryption is an imprecise one.

In general terms, it refers to encryption schemes that make it impractically difficult for unauthorized persons or systems to gain access to plaintext that has been encrypted.



Strong encryption should prevent unauthorized access regardless of the resources available to the attacker.



The strength of encryption depends on multiple factors that work together to create a robust security system that can withstand various types of attacks.

Strong Encryption Properties (2/2)



Algorithm

Appropriate choice of cryptographic algorithm



Key Length

Use of sufficiently long key lengths



Protocols

Appropriate choice of protocols



Implementation

Well-engineered implementation



Flaw-free

Absence of deliberately introduced hidden flaws

Brute-Force Attack Details (1/2)



A brute-force attack involves trying every possible key until an intelligible translation of the ciphertext into plaintext is obtained.



On average, half of all possible keys must be tried to achieve success.

That is, if there are X different keys, on average an attacker would discover the actual key after $X/2$ tries.



It is important to note that there is more to a brute-force attack than simply running through all possible keys.

Unless known plaintext is provided, the analyst must be able to recognize plaintext as plaintext.



If the message is just plain text in English, then the result pops out easily.

Brute-Force Attack Details (2/2)



If the text message has been compressed before encryption, then recognition is more difficult.

And if the message is some more general type of data, such as a numerical file, and this has been compressed, the problem becomes even more difficult to automate.



Thus, to supplement the brute-force approach, some degree of knowledge about the expected plaintext is needed.

Some means of automatically distinguishing plaintext from garble is also needed.



The brute-force attack becomes more challenging when dealing with encrypted and compressed data.

The attacker must develop sophisticated methods to identify the correct decryption result.



Public-Key Cryptography



Asymmetric Encryption Systems and Their Applications

Introduction to Public-Key Cryptography (1/2)



Symmetric encryption, also referred to as conventional encryption or single-key encryption, was the only type of encryption in use prior to the development of public-key encryption in the 1970s.



It remains by far the most widely used of the two types of encryption.



Part Two examines a number of symmetric ciphers.

In this chapter, we look briefly at a different approach known as public-key cryptography.



Public-key encryption was a breakthrough developed by Whitfield Diffie and Martin Hellman in 1976.

This approach addressed fundamental problems with conventional encryption.

Introduction to Public-Key Cryptography (2/2)



Diffie and Hellman achieved an astounding breakthrough by coming up with a method that addressed both problems.



The method was radically different from all previous approaches to cryptography.
It went back over four millennia.



In the next subsection, we look at the overall framework for public-key cryptography.
Then we examine the requirements for the encryption and decryption algorithm that is at the heart of the scheme.



Asymmetric algorithms rely on one key for encryption and a different but related key for decryption.

This fundamental characteristic enables the revolutionary approach to cryptographic security.

Public-Key Cryptosystem Ingredients (1/2)



A public-key encryption scheme has six ingredients.

1

The first ingredient is plaintext, which is the readable message or data that is fed into the algorithm as input.

2

The second ingredient is the encryption algorithm, which performs various transformations on the plaintext.

3

The third ingredient is public and private keys.

This is a pair of keys that have been selected so that if one is used for encryption, the other is used for decryption.

The exact transformations performed by the algorithm depend on the public or private key that is provided as input.

Public-Key Cryptosystem Ingredients (2/2)

4

The fourth ingredient is ciphertext, which is the encrypted message produced as output.

It depends on the plaintext and the key.

For a given message, two different keys will produce two different ciphertexts.

5

The fifth ingredient is the decryption algorithm.

This algorithm accepts the ciphertext and the matching key and produces the original plaintext.

6

The sixth ingredient is that either of the two related keys can be used for encryption, with the other used for decryption.

This flexibility enables both confidentiality and authentication applications.

Essential Steps of Public-Key Encryption (1/2)

1

Each user generates a pair of keys to be used for the encryption and decryption of messages.

2

Each user places one of the two keys in a public register or other accessible file.

This is the public key. The companion key is kept private.

As shown in the model, each user maintains a collection of public keys obtained from others.

3

If Bob wishes to send a confidential message to Alice, Bob encrypts the message using Alice's public key.

When Alice receives the message, she decrypts it using her private key.



No other recipient can decrypt the message because only Alice knows Alice's private key.

Essential Steps of Public-Key Encryption (2/2)



With this approach, all participants have access to public keys.

Private keys are generated locally by each participant and therefore need never be distributed.



As long as a user's private key remains protected and secret, incoming communication is secure.



At any time, a system can change its private key and publish the companion public key to replace its old public key.

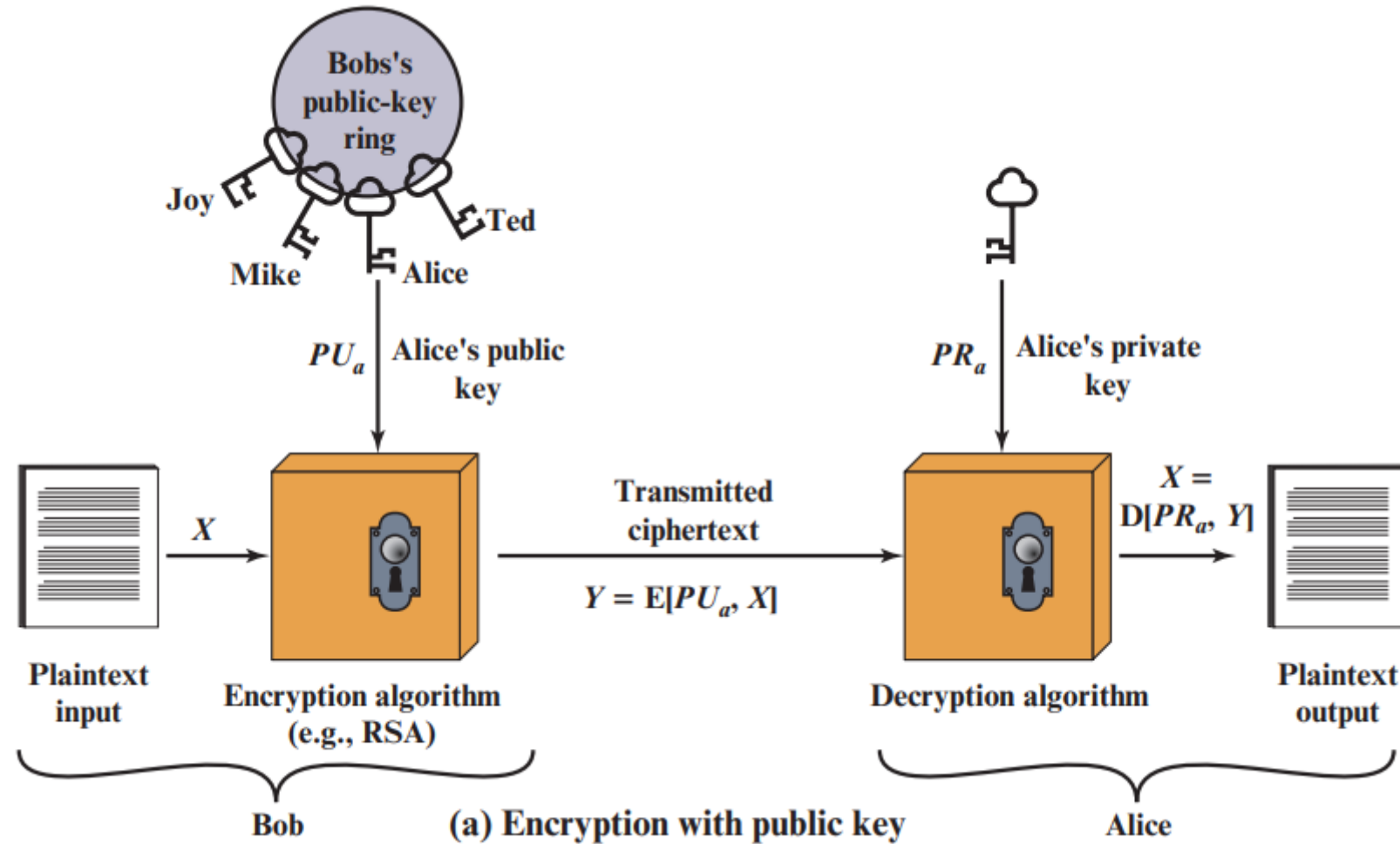
This provides flexibility and enhanced security.

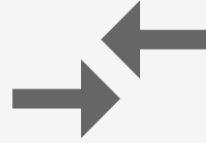


The key pair generation process must be carefully designed to ensure the mathematical relationship between the keys.

The security of the entire system depends on the proper management of both public and private keys.

Public-Key Cryptography





Comparison: Symmetric vs Public-Key

Understanding the Differences Between Encryption Approaches

Symmetric vs Public-Key Encryption (1/2)



To discriminate between the two types of encryption, we refer to the key used in symmetric encryption as a secret key.



The two keys used for asymmetric encryption are referred to as the public key and the private key.



Invariably, the private key is kept secret.

It is referred to as a private key rather than a secret key to avoid confusion with symmetric encryption.



Table 9.2 summarizes some of the important aspects of symmetric and public-key encryption.

This comparison highlights the fundamental differences between these two approaches.

Symmetric vs Public-Key Encryption (2/2)



Conventional Encryption

Needed to Work:

- Same algorithm with same key for encryption and decryption
- Sender and receiver must share algorithm and key

Needed for Security:

- Key must be kept secret
- Impossible or impractical to decipher if key is secret
- Algorithm + ciphertext samples insufficient to determine key



Public-Key Encryption

Needed to Work:

- One algorithm for encryption, related algorithm for decryption with key pair
- Sender and receiver each have one matched key

Needed for Security:

- One of two keys must be kept secret
- Impossible or impractical to decipher if one key is secret
- Algorithm + one key + ciphertext insufficient to determine other key

Conventional and Public-Key Encryption

Conventional Encryption	Public-Key Encryption
<i>Needed to Work:</i> 1. The same algorithm with the same key is used for encryption and decryption. 2. The sender and receiver must share the algorithm and the key. <i>Needed for Security:</i> 1. The key must be kept secret. 2. It must be impossible or at least impractical to decipher a message if the key is kept secret. 3. Knowledge of the algorithm plus samples of ciphertext must be insufficient to determine the key.	<i>Needed to Work:</i> 1. One algorithm is used for encryption and a related algorithm for decryption with a pair of keys, one for encryption and one for decryption. 2. The sender and receiver must each have one of the matched pair of keys (not the same one). <i>Needed for Security:</i> 1. One of the two keys must be kept secret. 2. It must be impossible or at least impractical to decipher a message if one of the keys is kept secret. 3. Knowledge of the algorithm plus one of the keys plus samples of ciphertext must be insufficient to determine the other key.

Authentication with Public-Key (1/2)



We mentioned earlier that either of the two related keys can be used for encryption.
The other key is used for decryption.



This enables a rather different cryptographic scheme to be implemented.

The scheme illustrated in Figure ahead shows the use of public-key encryption to provide authentication.



In this case, A prepares a message to B and encrypts it using A's private key before transmitting it.

B can decrypt the message using A's public key.

Because the message was encrypted using A's private key, only A could have prepared the message.

Therefore, the entire encrypted message serves as a digital signature.

Authentication with Public-Key (2/2)



In addition, it is impossible to alter the message without access to A's private key.
The message is authenticated both in terms of source and in terms of data integrity.



However, it is important to emphasize that the encryption process does not provide confidentiality.
The message being sent is safe from alteration but not from eavesdropping.

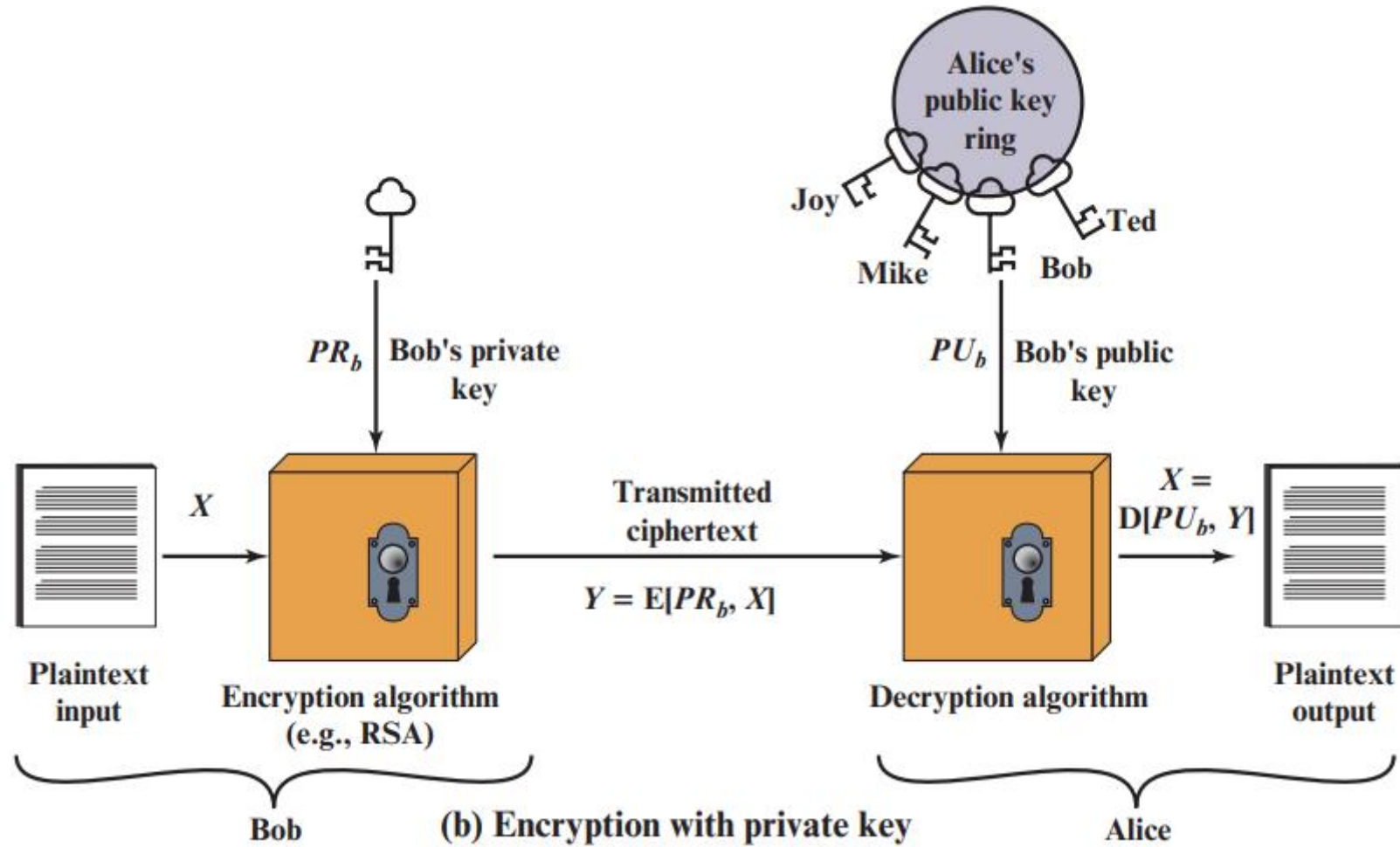


This is obvious in the case of a signature based on a portion of the message.
The rest of the message is transmitted in the clear.

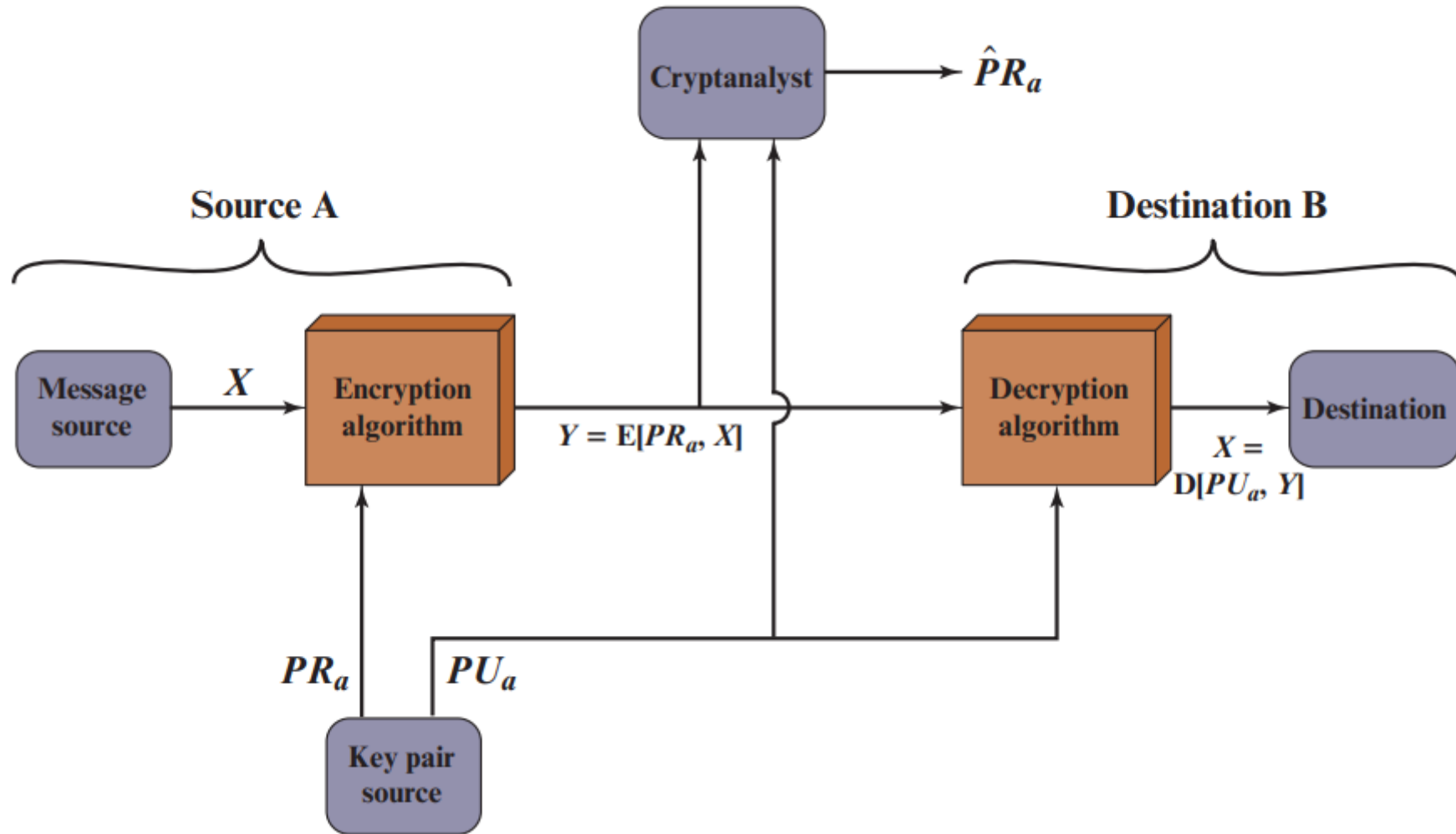


Even in the case of complete encryption, there is no protection of confidentiality.
Any observer can decrypt the message by using the sender's public key.

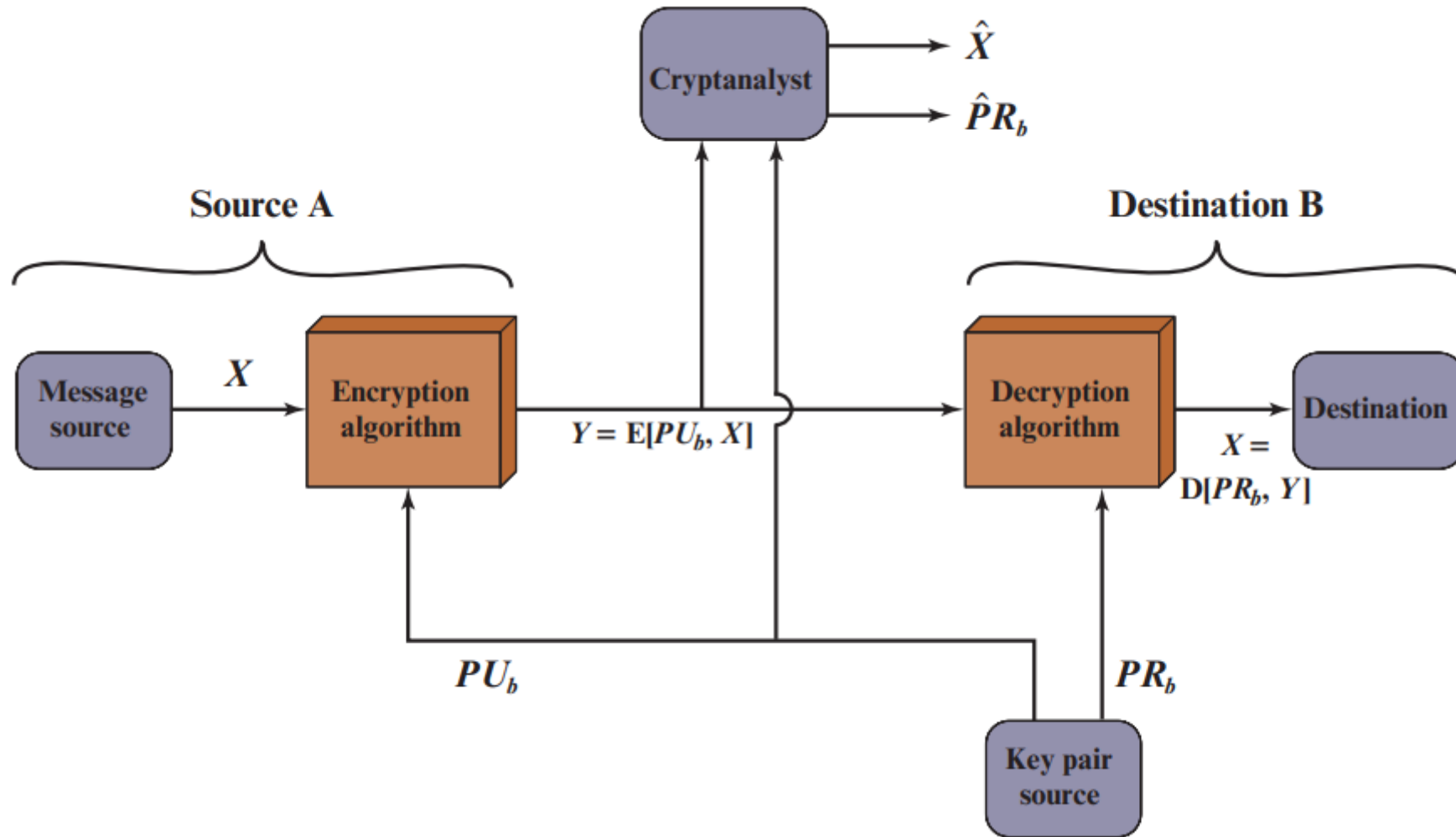
Public-Key Cryptography - Authentication



Public-Key Cryptography - Authentication



Public-Key Cryptography - Confidentiality



Combined Authentication and Secrecy (1/2)



It is possible to provide both the authentication function and confidentiality by a double use of the public-key scheme.



In this case, we begin as before by encrypting a message using the sender's private key.

This provides the digital signature.



Next, we encrypt again using the receiver's public key.

The final ciphertext can be decrypted only by the intended receiver who has the matching private key.

Thus, confidentiality is provided.



The disadvantage of this approach is that the public-key algorithm must be exercised four times rather than two in each communication, resulting in increased computational overhead.

Combined Authentication and Secrecy (2/2)



The equations for this combined approach are:

$$Z = E(\text{PUB}, E(\text{PRa}, X))$$

$$X = D(\text{PUa}, D(\text{PRb}, Z))$$



The first encryption with the sender's private key provides authentication.

The second encryption with the receiver's public key provides confidentiality.

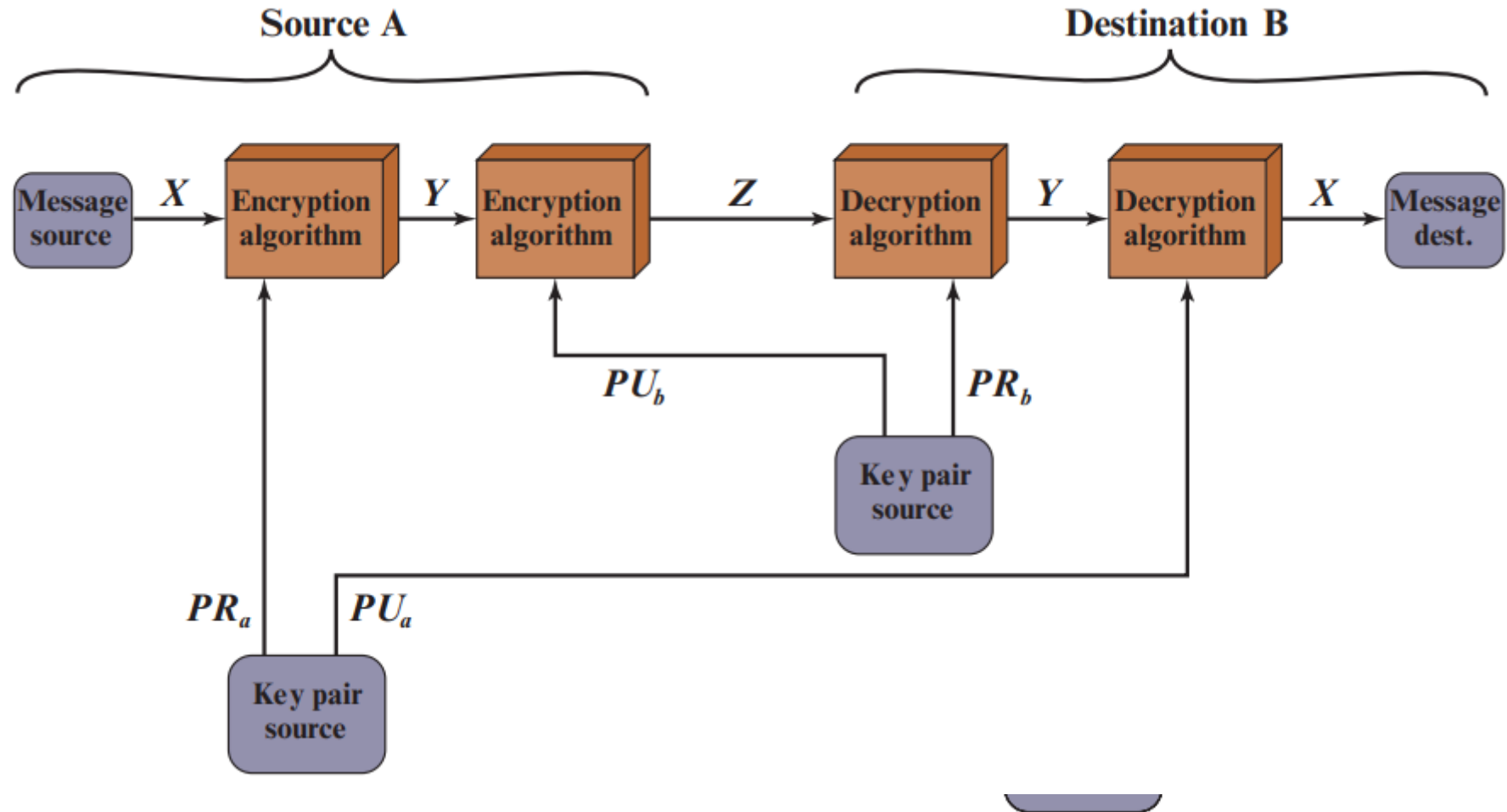


This double encryption approach ensures that both authentication and confidentiality are achieved.



However, the computational cost is higher due to performing four cryptographic operations instead of two.

Public-Key Cryptosystem: Authentication and Secrecy



Applications of Public-Key Cryptosystems (1/2)



Before proceeding, we need to clarify one aspect of public-key cryptosystems that is otherwise likely to lead to confusion.



Public-key systems are characterized by the use of a cryptographic algorithm with two keys.

One key is held private and one is available publicly.



Depending on the application, the sender uses either the sender's private key or the receiver's public key, or both.

The sender performs some type of cryptographic function.



In broad terms, we can classify the use of public-key cryptosystems into three categories.

These categories are encryption/decryption, digital signature, and key exchange.

Applications of Public-Key Cryptosystems (2/2)



The first category is **encryption/decryption**.

The sender encrypts a message with the recipient's public key, and the recipient decrypts the message with the recipient's private key.



The second category is **digital signature**.

The sender signs a message with its private key. Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data that is a function of the message.



The third category is **key exchange**.

Two sides cooperate to exchange a session key, which is a secret key for symmetric encryption generated for a particular transaction. This session key is valid for a short period of time.



Several different approaches are possible for key exchange, involving the private key(s) of one or both parties.

Requirements for Public-Key Cryptography (1/2)

i

The cryptosystem depends on a cryptographic algorithm based on two related keys.

Diffie and Hellman postulated this system without demonstrating that such algorithms exist, but they laid out the conditions that such algorithms must fulfill.

1

It is computationally easy for a party B to generate a key pair (public key P_{Ub} and private key PR_b).

2

It is computationally easy for a sender A, knowing the public key and the message to be encrypted, M , to generate the corresponding ciphertext $C = E(P_{Ub}, M)$.

3

It is computationally easy for the receiver B to decrypt the resulting ciphertext using the private key to recover the original message.

Requirements for Public-Key Cryptography (2/2)

- 4 It is computationally infeasible for an adversary, knowing the public key, PUB, to determine the private key, PRb.
- 5 It is computationally infeasible for an adversary, knowing the public key, PUB, and a ciphertext, C, to recover the original message, M.
- 6 The two keys can be applied in either order, meaning that $M = D[PUB, E(PRb, M)] = D[PRb, E(PUB, M)]$.



Formidable Requirements

Only a few algorithms have received widespread acceptance in the several decades since the concept of public-key cryptography was proposed.

Trap-Door One-Way Functions (1/2)



Before elaborating on why the requirements are so formidable, let us first recast them.

The requirements boil down to the need for a trap-door one-way function.



A one-way function maps a domain into a range such that every function value has a unique inverse.

The calculation of the function is easy, whereas the calculation of the inverse is infeasible.



$Y = f(X)$ is easy, but $X = f^{-1}(Y)$ is infeasible.



Easy is defined as a problem solvable in polynomial time as a function of input length.

For input length n , time is proportional to n^a (where a is a fixed constant), placing the algorithm in class P.

Trap-Door One-Way Functions (2/2)



The term infeasible is a much fuzzier concept.

A problem is infeasible if effort grows faster than polynomial time as a function of input size (e.g., time proportional to 2^n).



It is difficult to determine if a particular algorithm exhibits this complexity.

Traditional complexity measures (worst-case or average-case) are inadequate for cryptography.



A trap-door one-way function is easy in one direction and infeasible in the other unless additional information (k) is known.

Summary:

- $Y = f_k(X)$ is easy if k and X are known
- $X = f_k^{-1}(Y)$ is easy if k and Y are known
- $X = f_k^{-1}(Y)$ is infeasible if Y is known but k is not known

Public-Key Cryptanalysis (1/2)



As with symmetric encryption, a public-key encryption scheme is vulnerable to a brute-force attack.

The countermeasure is the same: Use large keys.



Public-key systems depend on invertible mathematical functions.

Function complexity may grow more rapidly than linearly with key size.



Key size must be large enough to make brute-force attack impractical but small enough for practical encryption and decryption.



Proposed key sizes make brute-force attack impractical but result in speeds too slow for general-purpose use.

Public-key encryption is currently confined to key management and signature applications.

Public-Key Cryptanalysis (2/2)



Another form of attack is to find some way to compute the private key given the public key.

It has not been mathematically proven that this attack is infeasible for a particular public-key algorithm.



Any given algorithm, including the widely used RSA algorithm, is suspect.

The history of cryptanalysis shows that problems can be solved if looked at from different perspectives.



A probable-message attack is peculiar to public-key systems.

If a message consists solely of a 56-bit DES key, an adversary could encrypt all possible keys using the public key and discover the encrypted key.



This attack can be thwarted by appending some random bits to such simple messages.



Acknowledgement

The material presented in these slides incorporates content from books, academic lecture notes, online resources, and generated text. All rights remain with the original authors and publishers. This content is used solely for educational purposes to illustrate and clarify concepts. No copyright infringement is intended.